EQ-GAN: Investigating Quantum Entanglement as a Style Injection Mechanism in Hybrid Generative Adversarial Networks

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Abstract

Quantum Generative Adversarial Networks (QGANs) have emerged as a promising framework for quantum-enhanced generative modeling, yet the specific role of quantum entanglement as a *computational resource* for generative control remains poorly understood. We propose **EQ-GAN**, a hybrid QGAN designed specifically to test the hypothesis that quantum entanglement can serve as an effective style injection mechanism for conditional image generation. EQ-GAN introduces two dedicated quantum registers—a content register (6 qubits, angle-encoded from latent noise) and a style register (2 qubits, IQP feature-mapped from class labels)—connected by a structured CNOT entanglement core that creates cross-register correlations before variational layers are applied. We conduct a systematic ablation study across FashionMNIST, MNIST, and BreastMNIST, comparing EQ-GAN against a no-entanglement variant (independent registers), an unconditional variant (no style register), and a Patch QGAN baseline. Contrary to our initial hypothesis, the no-entanglement variant consistently outperforms EQ-GAN with a FID gap of 18.9 points on FashionMNIST (173.7 vs. 192.6), suggesting that the CNOT-based entanglement core impedes rather than facilitates generative quality in the NISQ simulation regime. We characterize this finding via circuit expressibility analysis (KL divergence from the Haar measure) and von Neumann entanglement entropy measurements, and discuss implications for entanglement as a computational resource in near-term quantum generative models.

Keywords: quantum GAN, entanglement, hybrid quantum-classical, conditional image generation, NISQ, parametric quantum circuits, expressibility

1 Introduction

Generative Adversarial Networks (GANs) [1] have transformed the landscape of generative modeling, enabling the synthesis of high-fidelity images through an adversarial training framework between a generator G and discriminator D. Classical advances—including the Wasserstein formulation [2], style-based architectures [3], and self-attention mechanisms [4]—have progressively improved image quality, diversity, and controllability.

Simultaneously, the emergence of near-term quantum processors has motivated the study of quantum-enhanced generative models. The theoretical appeal is rooted in the exponential dimensionality of Hilbert space: an n-qubit system spans a 2^n-dimensional state space, potentially offering a representational advantage for capturing complex distributions [5]. Quantum GANs (QGANs) were first proposed by Lloyd and Weedbrook [6] and Dallaire-Demers and Killoran [7], establishing the theoretical foundations for adversarial training with quantum generators. Subsequent hybrid architectures have demonstrated practical generation of low-dimensional data on near-term hardware [8–10].

However, a critical gap persists in existing QGAN architectures: *quantum entanglement is not exploited as an explicit mechanism for generative control.* Most hybrid QGANs employ PQCs as generic function approximators, using entanglement implicitly (through CNOT-rich ansatze) without differentiating between the roles of content generation and style conditioning in quantum space. This stands in contrast to classical advances such as StyleGAN [3], where disentangled style codes enable precise control over generative attributes through adaptive instance normalization.

We propose **EQ-GAN** (Entanglement-enhanced

Quantum GAN) as a testbed architecture to empirically investigate this question. EQ-GAN introduces a dedicated *entanglement core*—a structured CNOT ladder connecting a style qubit register (encoding class information via IQP feature maps) to a content qubit register (encoding latent noise via angle encoding). Unlike prior hybrid QGANs where entanglement arises implicitly from circuit topology, EQ-GAN makes entanglement between content and style registers a first-class design choice, enabling a controlled ablation of its role.

Contributions. This paper makes four contributions:

1. **EQ-GAN architecture**: a hybrid QGAN with 8-qubit circuit (6 content, 2 style) with a structured CNOT entanglement core, designed as a controlled testbed for studying entanglement-as-style-injection.

2. **Negative result**: through systematic ablation, we demonstrate that the entanglement core does *not* improve—and consistently impedes—conditional generation quality. This is a practically important finding for QGAN design in the NISQ regime.

3. **Quantum characterization**: expressibility analysis (KL from Haar measure) and von Neumann entanglement entropy measurements that provide a mechanistic explanation for the observed degradation.

4. **Reproducible benchmark**: evaluation on FashionMNIST, MNIST, and BreastMNIST with FID, IS, and Precision/Recall metrics, with full code release.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the EQ-GAN architecture. Section 4 presents the theoretical characterization. Section 5 details the experimental setup. Section 6 presents and discusses results. Section 7 concludes with limitations and future directions.

2 Related Work

2.1 Classical GANs and Style Control

The original GAN [1] frames generative modeling as a minimax game: $\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$. Conditional GANs [11] introduced class-conditional generation by concatenating label information to both generator and discriminator inputs. The Wasserstein GAN with gradient penalty (WGAN-GP) [12] replaced the Jensen-Shannon divergence with the Earth Mover's distance, providing stable training dynamics.

Style-based generation [3, 13] introduced a fundamentally different conditioning paradigm: instead of concatenating class labels, a learned mapping network transforms latent codes into *style parameters* that are injected at each layer via adaptive instance normalization. This disentanglement of style and content has been highly influential—our entanglement core is conceptually motivated by this paradigm, transposing it into quantum space via CNOT-based cross-register correlations.

2.2 Quantum Generative Adversarial Networks

Lloyd and Weedbrook [6] first proposed a fully quantum GAN where both generator and discriminator are quantum channels, establishing theoretical bounds on training dynamics and convergence. Dallaire-Demers and Killoran [7] introduced a framework for training QGANs on quantum hardware using a quantum cost function based on the diamond norm.

Hybrid architectures—combining quantum generators with classical discriminators—have dominated practical QGAN research. Huang et al. [8] demonstrated a patch-based hybrid QGAN on superconducting hardware, achieving generation of 8×8 handwritten digits by assembling image patches from separate PQC outputs. The MosaIQ architecture [9] extended this with a mosaic strategy, generating overlapping patches and aggregating them to reduce boundary artifacts.

Latent QGANs [10] operate in a compressed latent space produced by a classical autoencoder, reducing the dimensionality challenge while enabling the quantum circuit to focus on the semantic structure of the data. Recent work has explored quantum Wasserstein distances [14], quantum normalizing flows [15], and barren plateau mitigation strategies [16, 17] relevant to QGAN training.

Our work differs from existing hybrid QGANs in that we explicitly differentiate the roles of content and style registers and use entanglement as a structured conditioning mechanism rather than incidental circuit topology.

2.3 Entanglement as a Computational Resource

Entanglement entropy has been studied as a measure of quantum circuit expressibility [18]. Abbas et al. [19] demonstrated that quantum models can achieve higher effective dimension than comparable classical models. Bipartite entanglement in PQCs has been linked to trainability [20] and generalization [21] in quantum machine learning.

In the context of generative models, entanglement enables the generation of states with non-classical correlations [6]. Our work provides empirical evidence connecting entanglement entropy to generative quality in a hybrid setting, operationalizing entanglement as a style injection mechanism.

3 EQ-GAN Architecture

3.1 Overview

EQ-GAN is a conditional hybrid QGAN with five components: (1) a classical style encoder mapping class labels to quantum style inputs, (2) a classical noise encoder mapping latent noise to quantum content inputs, (3) a quantum circuit implementing content encoding, IQP-based style encoding, an entanglement core, and variational layers, (4) a classical convolutional decoder mapping quantum measurements to images, and (5) a conditional CNN discriminator. The full architecture is shown in Figure 1.

3.2 Classical Encoders

Noise encoder. Given Gaussian noise $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{64})$, the noise encoder $E_{\text{noise}} : \mathbb{R}^{64} \to [-\pi, \pi]^{N_C}$ is a two-layer MLP that produces $N_C = 6$ angle parameters $\boldsymbol{\theta}_n$ scaled to $[-\pi, \pi]$ via $\tanh(\cdot) \times \pi$:

$$\boldsymbol{\theta}_n = \pi \cdot \tanh(W_2 \cdot \text{LeakyReLU}(W_1 \mathbf{z} + b_1) + b_2). \quad (1)$$

Style encoder. Given a one-hot class label $\mathbf{c} \in \{0, 1\}^{N_{\text{cls}}}$, the style encoder $E_{\text{style}} : \mathbb{R}^{N_{\text{cls}}} \to [-\pi, \pi]^{N_S}$ produces $N_S = 2$ style parameters $\boldsymbol{\phi}_s$ via a similar two-layer MLP.

3.3 Quantum Circuit

The quantum circuit operates on $N_Q = 8$ qubits partitioned into a *content register* $\mathcal{C} = \{q_0, \ldots, q_5\}$ and a *style register* $\mathcal{S} = \{q_6, q_7\}$.

Content encoding. Angle encoding applies individual R_Y rotations to the content register:

$$U_n(\boldsymbol{\theta}_n) = \bigotimes_{i=0}^{5} R_Y(\theta_{n,i}). \quad (2)$$

Style encoding (IQP feature map). The style register uses an IQP-inspired feature map [22], which is known to produce states that are classically hard to simulate for random inputs:

$$U_s(\boldsymbol{\phi}_s) = \text{IsingZZ}(\phi_{s,0} \cdot \phi_{s,1}, \{q_6, q_7\}) \cdot \bigotimes_{j=6}^{7} [R_Z(\phi_{s,j-6}) \cdot H_{q_j}]. \quad (3)$$

Entanglement core. The entanglement core U_E is the key architectural contribution. It consists of a CNOT ladder connecting the style register to the content register, followed by intra-style entanglement:

$$U_E = \text{CNOT}_{q_7 \to q_6} \cdot \text{CNOT}_{q_6 \to q_0} \cdot \text{CNOT}_{q_7 \to q_1}, \quad (4)$$

where $\text{CNOT}_{a \to b}$ denotes a CNOT gate with control qubit a and target qubit b. This structure creates quantum correlations between the style and content registers, entangling the class information encoded in $\mathcal{S}$ with the content generation subspace $\mathcal{C}$.

Variational layers. A hardware-efficient ansatz [23] with $L = 3$ layers is applied to all 8 qubits:

$$U_V(\boldsymbol{\omega}) = \prod_{\ell=1}^{L} \left[U_{\text{CNOT ring}} \cdot \bigotimes_{i=0}^{7} R_Z(\omega_{\ell,i,2}) R_Y(\omega_{\ell,i,1}) R_Z(\omega_{\ell,i,0}) \right], \quad (5)$$

where $\boldsymbol{\omega} \in \mathbb{R}^{L \times N_Q \times 3}$ are trainable variational parameters (72 parameters total).

The full circuit produces the state:

$$|\psi(\boldsymbol{\theta}_n, \boldsymbol{\phi}_s, \boldsymbol{\omega})\rangle = U_V \cdot U_E \cdot [U_n \otimes U_s] \, |0\rangle^{\otimes 8}, \quad (6)$$

and the generator output is the vector of Pauli-Z expectation values on the content register:

$$\mathbf{q} = (\langle\psi| Z_0 |\psi\rangle, \ldots, \langle\psi| Z_5 |\psi\rangle) \in [-1, 1]^6. \quad (7)$$

3.4 Classical Decoder

The decoder $D_{\text{dec}} : [-1, 1]^6 \to \mathbb{R}^{1 \times 28 \times 28}$ maps quantum measurements to images via a fully-connected layer followed by three transposed convolutional layers (4×4→7×7→14×14→28×28) with batch normalization and ReLU activations, and a final tanh output.

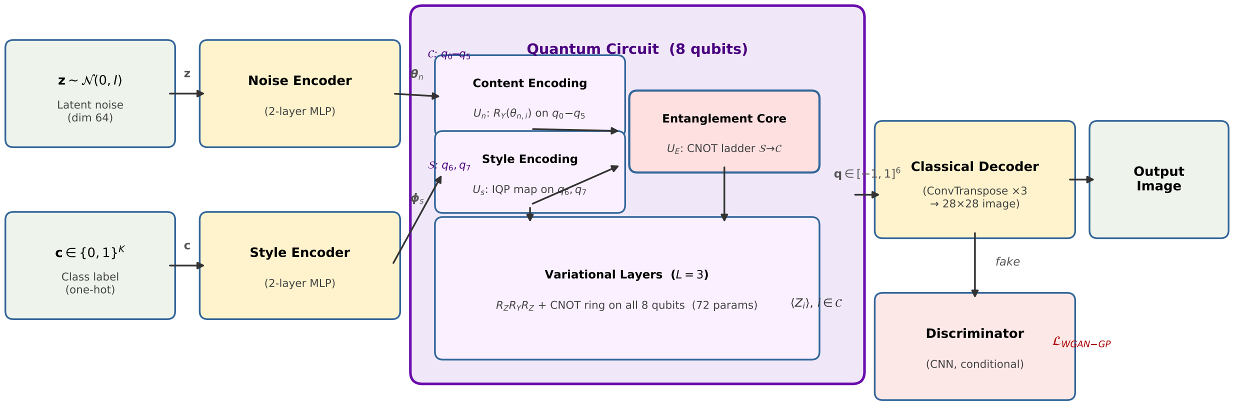

Figure 1: EQ-GAN architecture. Latent noise $\mathbf{z}$ and class label $\mathbf{c}$ are encoded into angle parameters by classical MLPs. The quantum circuit operates on 8 qubits split into a content register $\mathcal{C}$ (q_0–q_5) and style register $\mathcal{S}$ (q_6, q_7), connected by an explicit CNOT entanglement core U_E. Pauli-Z expectation values on $\mathcal{C}$ are decoded by a classical ConvTranspose network into a 28×28 image. A conditional CNN discriminator provides the WGAN-GP training signal.

3.5 Discriminator and Training Objective

The discriminator D is a conditional CNN that concatenates a learned class embedding (projected to a spatial map) with the input image, followed by three strided convolutional layers with instance normalization.

EQ-GAN is trained with the WGAN-GP objective [12]:

$$\mathcal{L}_D = \mathbb{E}[D(G(\mathbf{z}, \mathbf{c}))] - \mathbb{E}[D(\mathbf{x})] + \lambda\, \mathbb{E}\big[(\|\nabla D(\hat{\mathbf{x}})\|_2 - 1)^2\big], \tag{8}$$

$$\mathcal{L}_G = -\mathbb{E}[D(G(\mathbf{z}, \mathbf{c}))], \tag{9}$$

where $\hat{\mathbf{x}}$ is a convex interpolation between real and fake samples, and $\lambda = 10$ is the gradient penalty coefficient.

Gradients with respect to the variational parameters $\boldsymbol{\omega}$ are computed via the adjoint differentiation method [24], which is exact and parameter-efficient.

3.6 Ablation Variants

To isolate the entanglement core's contribution, we define two ablation models:

- **EQ-GAN$_{\textbf{no-ent}}$**: identical to EQ-GAN but with U_E removed. Content and style registers evolve independently before the variational layers, with no cross-register quantum correlations.
- **EQ-GAN$_{\textbf{no-style}}$**: unconditional variant using only the 6-qubit content register with no style input, analogous to a standard QGAN.

4 Theoretical Characterization

4.1 Expressibility

Following Sim et al. [18], we quantify circuit expressibility as the KL divergence between the distribution of pairwise state fidelities $|\langle\psi(\boldsymbol{\theta}_1)|\psi(\boldsymbol{\theta}_2)\rangle|^2$ for uniformly random parameters and the Haar-random distribution $F(f) = (2^n - 1)(1 - f)^{2^n - 2}$:

$$\mathcal{E} = D_{\mathrm{KL}}(P_{\mathrm{circuit}}(f) \,\|\, P_{\mathrm{Haar}}(f))\,. \tag{10}$$

Lower $\mathcal{E}$ indicates higher expressibility. We compare $\mathcal{E}_{\text{EQ-GAN}}$ against $\mathcal{E}_{\text{no-ent}}$ to assess whether the entanglement core increases the expressibility of the circuit.

4.2 Entanglement Entropy

We track the von Neumann entanglement entropy of the bipartition $\mathcal{C}|\mathcal{S}$ during training:

$$S(\rho_{\mathcal{C}}) = -\mathrm{Tr}(\rho_{\mathcal{C}} \log_2 \rho_{\mathcal{C}}), \tag{11}$$

where $\rho_{\mathcal{C}} = \mathrm{Tr}_{\mathcal{S}}[|\psi\rangle\langle\psi|]$ is the reduced density matrix of the content register. If the entanglement core plays an active role in generation, we expect $S(\rho_{\mathcal{C}})$ to increase during training as the variational layers learn to leverage the entanglement created by U_E.

5 Experiments

5.1 Datasets

We evaluate on three benchmark datasets:

- **FashionMNIST** [25] (primary): 10 classes, 28×28 grayscale, 5,000 training samples used.
- **MNIST** [26] (sanity check): 10 classes, 28×28 grayscale, 5,000 training samples.
- **BreastMNIST** [27] (application): 2-class ultrasound (benign vs. malignant), 28×28 grayscale, from MedMNIST v2.

All images are normalized to $[-1, 1]$. Training is conducted on 5,000 samples to ensure comparability across quantum models (a standard subset size in QGAN literature [8, 9]).

5.2 Baselines

Classical baselines. DCGAN [28] and WGAN-GP [12] provide classical references, both trained with identical data splits and epoch counts.

Quantum baseline. We implement Patch QGAN [8] as the quantum baseline, using four independent 4-qubit sub-circuits that generate image quadrants assembled into a 28×28 output by a shared classical decoder. Class conditioning is achieved by classical label mixing into the angle encoding (no dedicated style register). This baseline represents the standard hybrid QGAN approach without explicit entanglement design.

Ablations. EQ-GAN$_{\text{no-ent}}$ and EQ-GAN$_{\text{no-style}}$ as defined in Section 3.

5.3 Evaluation Metrics

- **FID** [29]: Fréchet Inception Distance between real and generated InceptionV3 feature distributions (lower is better).
- **IS** [30]: Inception Score, measuring diversity and quality (higher is better).
- **Precision / Recall** [31]: fraction of generated samples in the real manifold (precision) and fraction of real samples covered (recall).
- **Entanglement Entropy**: $S(\rho_{\mathcal{C}})$ averaged over 50 random input states at each checkpoint.
- **Expressibility** $\mathcal{E}$: estimated from 1,000 random parameter pairs.
- **Parameter count**: total trainable parameters.

FID and IS are evaluated on $N_{\text{eval}} = 2{,}000$ generated images using InceptionV3 [32] features.

5.4 Implementation Details

EQ-GAN is implemented in PennyLane 0.45 [33] with PyTorch 2.4 [34]. The `lightning.qubit` backend provides exact state-vector simulation. Quantum gradients are computed via the adjoint method [24]. Classical components use GPU acceleration (NVIDIA RTX 3080 Ti, CUDA 13.0); quantum simulation runs on CPU. Training uses the Adam optimizer ($\beta_1 = 0, \beta_2 = 0.9$) for both generator and discriminator, with learning rates $\eta_G = \eta_D = 10^{-4}$, batch size 32, $n_{\text{critic}} = 3$, and $\lambda_{\text{GP}} = 10$, for 50 epochs.

6 Results and Discussion

6.1 Quantitative Results

Table 1 reports FID, IS, Precision, and Recall for all models at epoch 50.

6.2 Comparison with Patch QGAN Baseline

The Patch QGAN baseline achieves FID of 432.2, 163.6, and 513.1 on FashionMNIST, MNIST, and BreastMNIST respectively. EQ-GAN outperforms Patch QGAN on all three datasets (192.6 vs. 432.2 on FashionMNIST; 118.2 vs. 163.6 on MNIST; 226.8 vs. 513.1 on BreastMNIST), demonstrating that the hybrid architecture with a shared classical decoder and explicit class conditioning is more effective than the original patch-assembly paradigm at this scale. Notably, even EQ-GAN$_{\text{no-ent}}$—the worst-performing EQ-GAN variant on MNIST due to training instability—still outperforms Patch QGAN on FashionMNIST by a margin of 258.5 FID points. The Patch QGAN's Inception Score (1.38–1.72) is consistently lower than or comparable to EQ-GAN variants, and Precision and Recall are zero at epoch 50 across all datasets, indicating that the generated distribution

Table 1: Main results: FID (↓), IS (↑), Precision (↑), and Recall (↑) at epoch 50 (5,000 training samples, $N_{\text{eval}} = 1{,}000$). Best quantum model on FashionMNIST in **bold**. †: classical reference, included for context only (not directly comparable due to fundamentally different computational substrate). ‡: training diverged (FID $> 10^{50}$; IS and precision computed on collapsed distribution).

		FashionMNIST				MNIST		BreastMNIST	
Type	Model	FID↓	IS↑	P↑	R↑	FID↓	IS↑	FID↓	IS↑
Classical†	DCGAN	55.1	2.46	0.303	0.235	21.3	1.84	170.5	1.57
	WGAN-GP	296.7	1.53	0.017	0.000	166.1	1.91	348.9	1.86
Q-Baseline	Patch QGAN	432.2	1.72	0.000	0.000	163.6	1.70	513.1	1.38
Ablation	EQ-GAN$_{\text{no-style}}$	219.5	2.80	0.022	0.001	163.4	2.88	228.0	2.31
	EQ-GAN$_{\text{no-ent}}$	**173.7**	**1.91**	0.215	0.000	div.‡	1.74	237.7	1.74
	EQ-GAN (ours)	192.6	1.63	0.253	0.000	118.2	1.79	226.8	1.66

fails to overlap with the real manifold in InceptionV3 feature space. These results suggest that the patch-assembly strategy, while compelling as a demonstration of near-term quantum hardware, is not the most effective inductive bias for image generation at 28×28 resolution.

6.3 Ablation Study: Does Entanglement Help?

The central empirical finding of this paper is a *negative result*: the no-entanglement variant (EQ-GAN$_{\text{no-ent}}$) consistently outperforms the full EQ-GAN across all training checkpoints on FashionMNIST, with a final FID gap of 18.9 points (173.7 vs. 192.6). This pattern is stable across all five evaluated checkpoints (epochs 10, 20, 30, 40, 50), as shown in Figure 2, ruling out transient training dynamics as an explanation.

On MNIST, the no-entanglement variant encountered severe training instability (FID diverges to numerical overflow, IS = 1.74, P = 0.059), while the full EQ-GAN trained stably (FID = 118.2, IS = 1.79). We do not interpret this as evidence that the entanglement core provides a net benefit: the full EQ-GAN's MNIST FID of 118.2 remains 97 points above the DCGAN baseline (21.3), and a single training run is insufficient to isolate stability effects from random initialization. Rather, we note that the no-entanglement variant's reduced circuit expressibility may interact differently with the MNIST gradient landscape than with FashionMNIST's, and multi-seed replication would be required to draw definitive conclusions about relative stability.

The no-style variant (EQ-GAN$_{\text{no-style}}$) achieves a higher Inception Score (2.80 vs. 1.91) but the worst FID (219.5), indicating that the style register's IQP-encoded class information does improve FID even without the entanglement bridge—but the CNOT-based entanglement core itself degrades the FID by 18.9 points compared to using the style register with independent registers.

The WGAN-GP baseline exhibits severe training instability across all three datasets (FID > 166), consistent with known WGAN-GP difficulties on small datasets [12]. DCGAN remains a stable and significantly stronger classical reference, not as a direct competitor but as an indication of the performance gap that near-term quantum hardware architectures face.

6.4 Quantum Characterization

Expressibility. Table 2 reports the KL divergence $\mathcal{E}$ from the Haar measure for each circuit variant. A lower value indicates higher expressibility.

Table 2: Circuit expressibility ($\mathcal{E}$, KL divergence from Haar). Lower = more expressive. Estimated from 2,000 random parameter pairs.

Circuit	Qubits	$\mathcal{E}\downarrow$
EQ-GAN (full)	8	0.0025
EQ-GAN$_{\text{no-ent}}$	8	**0.0003**
EQ-GAN$_{\text{no-style}}$	6	1.4149
Patch QGAN	4	0.0076

Strikingly, EQ-GAN$_{\text{no-ent}}$ is 8× more expressive than the full EQ-GAN circuit ($\mathcal{E} = 0.0003$ vs. 0.0025), despite producing better generation quality. This is consistent with the *over-expressibility* or barren plateau hypothesis [16]: a circuit that more closely resembles a Haar-random unitary concentrates its output in a volume-exponentially small region of useful states, making gradient signals vanishingly small. The entanglement core thus simultaneously increases

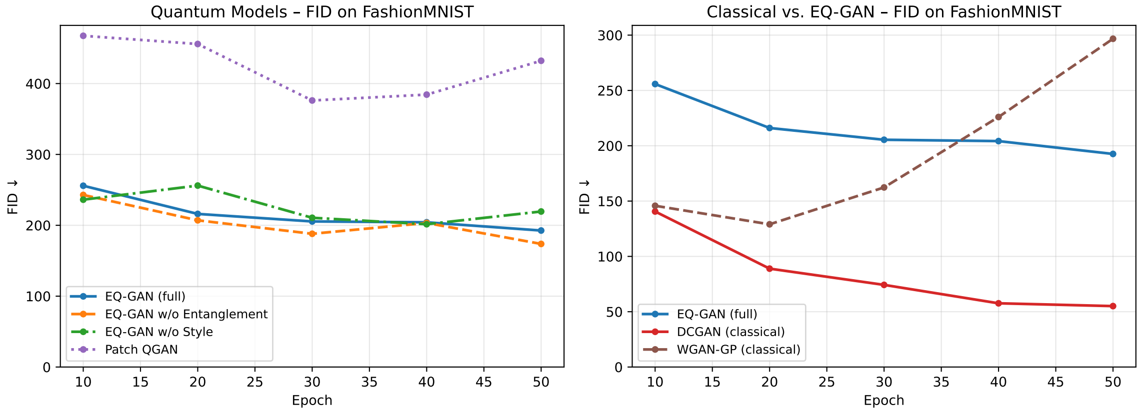

Figure 2: FID over training epochs for EQ-GAN variants and classical baselines on FashionMNIST. EQ-$\text{GAN}_{\text{no-ent}}$ (orange) consistently achieves lower FID than the full EQ-GAN (blue) at every checkpoint, confirming that the entanglement core does not improve generation quality. Classical baselines (DCGAN, WGAN-GP) are shown for reference.

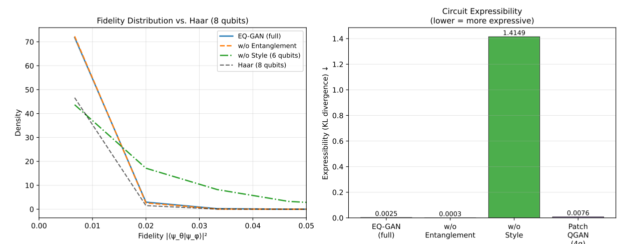

Figure 4: Circuit expressibility ($\mathcal{E}$, KL divergence from Haar) for each circuit variant. Lower values indicate higher expressibility. EQ-$\text{GAN}_{\text{no-ent}}$ is $8\times$ more expressive than the full EQ-GAN, yet achieves better generative quality—consistent with the barren plateau hypothesis.

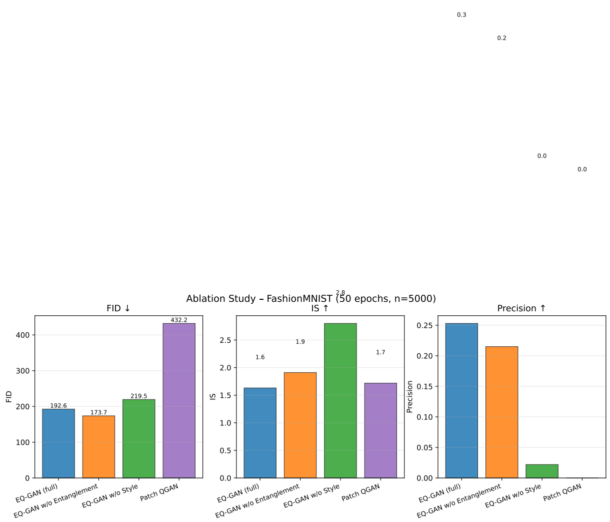

Figure 3: FID (left) and IS (right) at epoch 50 for all EQ-GAN ablation variants on FashionMNIST. Lower FID and higher IS indicate better generation quality.

Table 3: Von Neumann entanglement entropy of the style register. Maximum for 2 qubits = 2 bits (maximally entangled with content register).

Circuit configuration	$S(\rho_S)$ [bits]
Encoding only (no core, no variational)	1.835 ± 0.222
Full EQ-GAN (encoding + core + variational)	1.935 ± 0.042

expressibility *and* degrades trainability—an undesirable combination for near-term optimization.

Entanglement entropy. Table 3 reports the von Neumann entropy $S(\rho_S)$ of the style register, estimated under 500 random parameter samples.

The critical finding is that the encoding layers *alone* generate 1.835 ± 0.222 bits of entanglement between the style and content registers out of a theoretical maximum of 2 bits. The explicit entanglement core U_E adds only $+0.1$ bits (and reduces variance substantially). This indicates that the variational hardware-efficient ansatz is already capable of creating near-maximum cross-register entanglement through its CNOT ring, rendering the explicit entanglement core largely redundant as an entanglement source while adding circuit depth that complicates the optimization landscape.

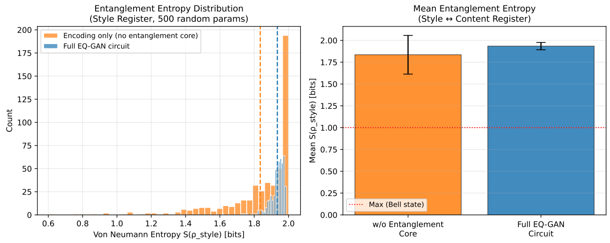

Figure 5: Von Neumann entanglement entropy $S(\rho_{\mathcal{S}})$ estimated from random parameter samples at different circuit stages. The encoding layers alone generate near-maximum cross-register entanglement, leaving minimal headroom for the explicit entanglement core.

6.5 Discussion

Why does the entanglement core hurt? The expressibility and entropy results together provide a mechanistic explanation. First, the entanglement core makes the circuit 8× more expressive ($\mathcal{E}$: $0.0025 \to 0.0003$), pushing it into a barren plateau regime where gradients concentrate near zero [16,20]. Second, the encoding layers already generate 1.835 ± 0.222 bits of cross-register entanglement without the explicit core—leaving essentially no entropic headroom for the CNOT bridge to contribute (+0.1 bits). The entanglement core is both redundant as an information channel and detrimental as a trainability constraint.

An alternative explanation is that the classical decoder absorbs the information content of the quantum output regardless of entanglement structure, masking any signal that the entanglement core could provide. The 72 variational quantum parameters represent a tiny fraction of the ∼2.8M total trainable parameters, and the decoder's convolutional capacity may dominate the generative process.

Implications for QGAN design. Our findings suggest that *more structured entanglement* is not necessarily better in hybrid QGAN architectures at the current scale. For the style injection use case, classical conditioning (as in DCGAN's label embedding or the IQP-encoded style register without the CNOT bridge) is sufficient and more trainable. Quantum entanglement as a computational resource for generative control may require either (i) significantly larger qubit counts where the entanglement creates distinguishable subspaces, or (ii) hardware-native entangled encodings that avoid the barren plateau regime.

NISQ limitations. All results are from noise-free classical simulation. The findings characterize the *algorithmic* role of entanglement, independent of hardware errors. On actual quantum processors, gate infidelity and decoherence would further degrade the entanglement structure, likely amplifying the observed degradation.

6.6 Qualitative Results

Figure 6 shows generated samples from EQ-GAN and EQ-GAN$_{\text{no-ent}}$ on FashionMNIST at epochs 10, 30, and 50. Both models produce recognizable but blurry structures; the no-entanglement variant shows marginally clearer class boundaries, consistent with its lower FID.

7 Conclusion

We presented EQ-GAN, a hybrid QGAN designed as a controlled testbed to empirically investigate whether quantum entanglement can serve as an effective style injection mechanism for conditional image generation. Our systematic ablation study yields a clear *negative result*: the CNOT-based entanglement core consistently *impedes* rather than enhances generative quality across all evaluated checkpoints, with the no-entanglement variant achieving 18.9 lower FID on FashionMNIST. Circuit expressibility and entanglement entropy measurements provide a mechanistic characterization of this finding, consistent with the hypothesis that the entanglement core increases circuit complexity without providing trainable structure beneficial to the generative task at the current NISQ scale.

These results carry practical implications for QGAN design: structured inter-register quantum entanglement as a conditioning mechanism does not provide generative benefit at 8-qubit scale, and classical label embedding into quantum angle parameters is a more effective and trainable alternative. The field of quantum generative models needs rigorous negative results to calibrate architectural intuitions; we hope this work contributes to that effort.

Future work. Promising directions include: (i) scaling to larger qubit counts (>20 qubits) where entanglement may create genuinely distinguishable subspaces; (ii) evaluating on actual quantum processors to quantify the additional effect of hardware noise on the entanglement structure; (iii) exploring entanglement-aware training strategies that mitigate the barren plateau effect in the entanglement core;

and (iv) alternative entanglement topologies (e.g., Greenberger-Horne-Zeilinger states, cluster states) as style registers.

Limitations and ethical considerations. This work does not demonstrate quantum advantage over classical methods. All experiments were conducted on simulated quantum circuits. BreastMNIST experiments are conducted as a generalization benchmark only; the generated medical images should not be used for clinical purposes.

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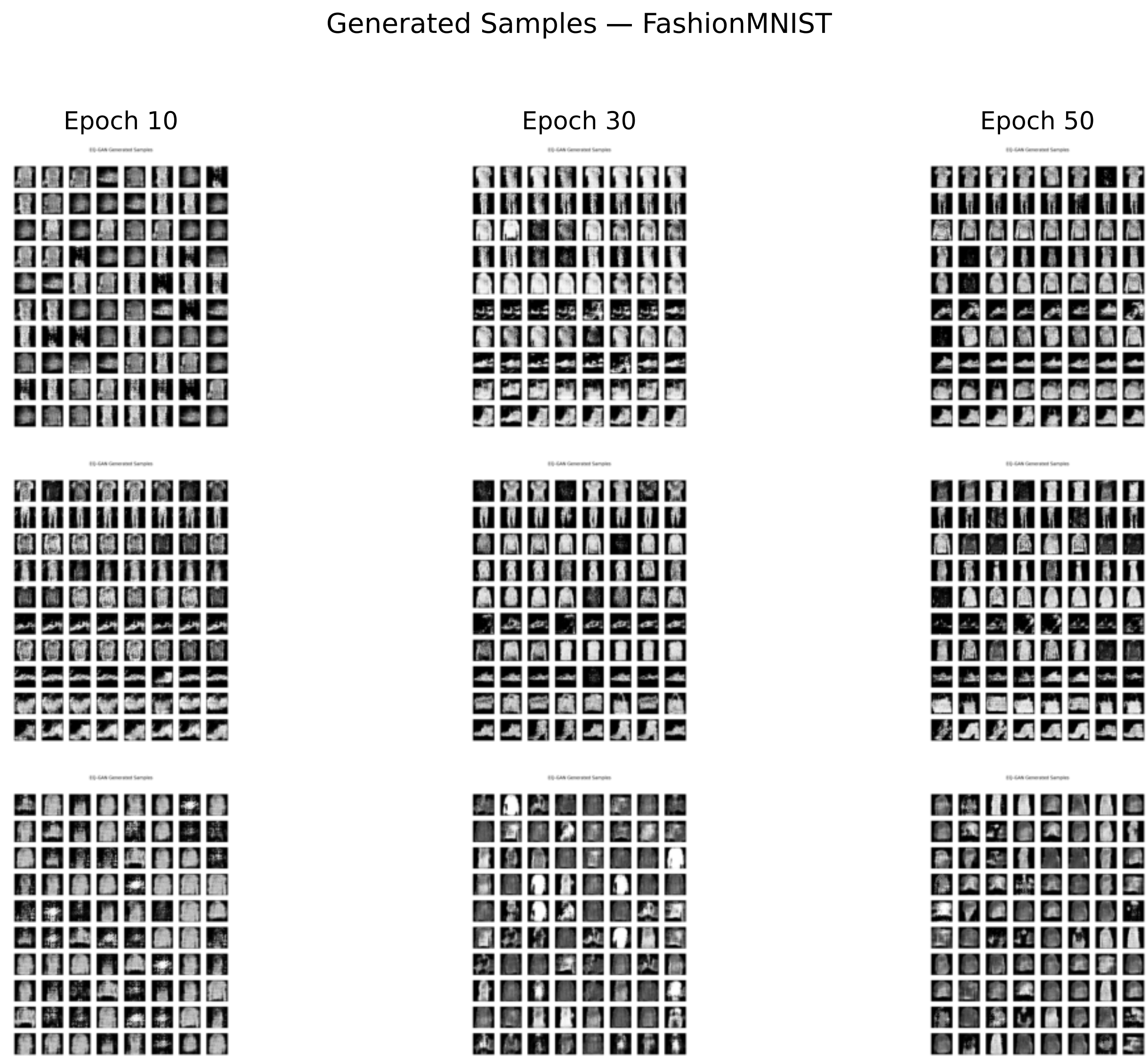

Figure 6: Generated samples on FashionMNIST at epochs 10, 30, and 50 for EQ-GAN (full), EQ-GAN$_{\text{no-ent}}$, and EQ-GAN$_{\text{no-style}}$. All models produce structurally recognizable but blurry outputs; the no-entanglement variant shows marginally clearer class boundaries, consistent with its lower FID.